# The transhipment problem

## The assignment problem

Michel Bierlaire

Optimization: principles and algorithms

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

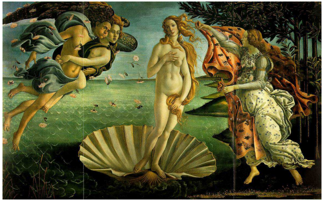
# The assignment problem

*The assignment problem consists in assignment $n$ tasks to $n$ resources at minimal cost.*

# Heirs

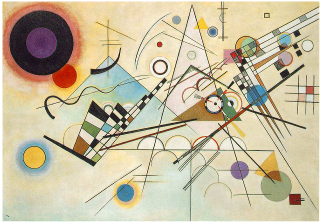
*After the death of her husband,*
*my grandmother has discovered four masterpieces in her attic*
*She does not want to keep them,*
*She would like to sell each of them to one of her four children.*
*Each child made an offer for the masterpieces of interest*

# Heirs

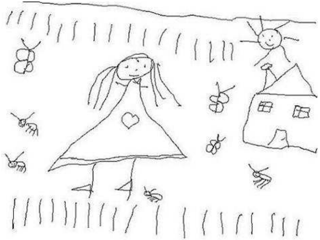


Botticelli, 1485

Bruegel, 1558



Kandinsky, 1923



Bierlaire, 1971

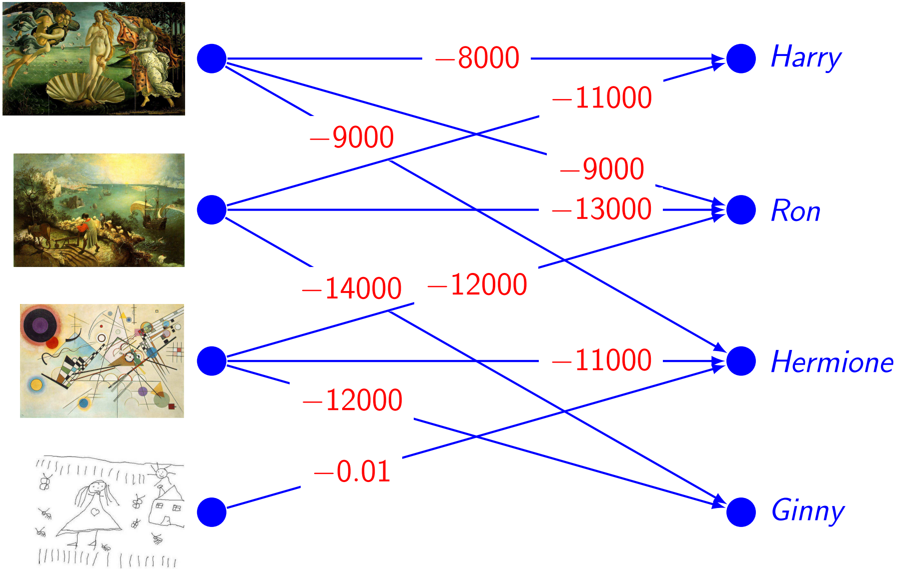
# Offers

*She wants to sell exactly one masterpiece to each child.*
*Which masterpiece should she sell to which child, to maximize her revenues?*

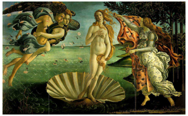
# Offers (kEuros)

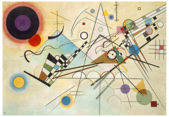
| | Botticelli | Bruegel | Kandinsky | Bierlaire |
|---|---|---|---|---|
| Harry | 8000 | 11000 | — | — |
| Ron | 9000 | 13000 | 12000 | — |
| Hermione | 9000 | — | 11000 | 0.01 |
| Ginny | — | 14000 | 12000 | — |

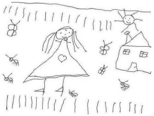
# Network


−8000
Harry
−11000
−9000
−9000
Ron
−13000
−14000
−12000
Hermione
−11000
−12000
−0.01
Ginny

# Network







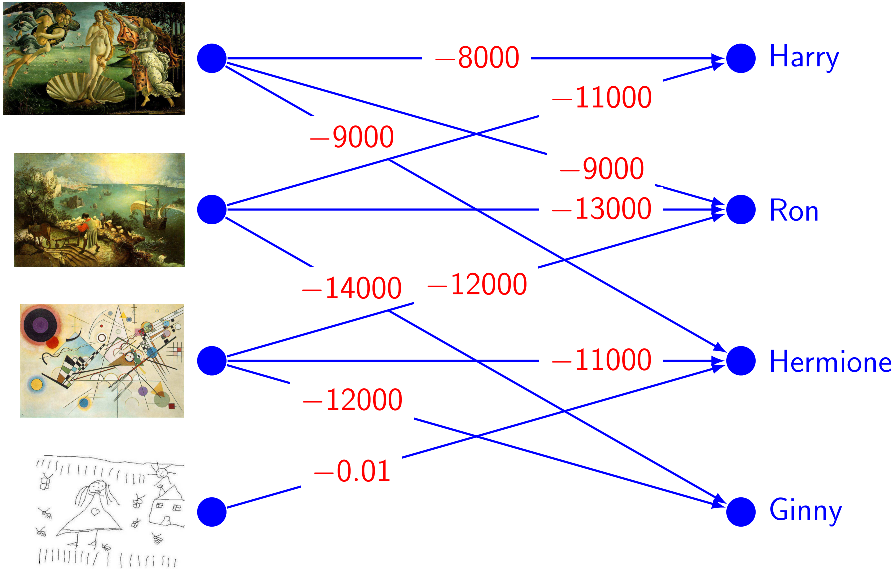
Harry

Ron

Hermione

Ginny

# Network


−8000
Harry
−11000
−9000
−9000
Ron
−13000
−14000
−12000
−11000
Hermione
−12000
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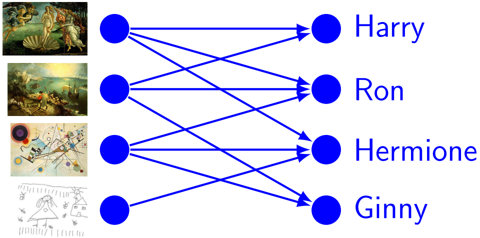
# Transhipment




- cost on each arc: $c_{ij} = -p_{ij}$
- supply for each resource node: 1
- supply for each task node: $-1$
- lower bound on each arc: 0
- upper bound on each arc: 1

# Transhipment

$$\min_{x \in \mathbb{R}^{n^2}} \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$$

*subject to*

$$\sum_{j=1}^{n} x_{ij} = 1, \qquad i = 1, \dots, n, \tag{1}$$

$$\sum_{i=1}^{n} x_{ij} = 1, \qquad j = 1, \dots, n, \tag{2}$$

$$x_{ij} \geq 0, \qquad i = 1, \dots, n,\ j = 1, \dots, n, \tag{3}$$

$$x_{ij} \leq 1, \qquad i = 1, \dots, n,\ j = 1, \dots, n. \tag{4}$$

*The variable $x_{ij}$ is to take on the value 1 if resource $i$ is assigned to task $j$, and 0 otherwise.*

# Transhipment

# Summary

*The assignment problem can be seen as a transportation problem where each supplier sends one unit of flow, and each customer receives one.*
*Even if there is no underlying network structure, it can be modeled as a transhipment problem*
*A conceptual network is representing the problem.*
*Again, it inherits the properties of the transhipment problem, such as total unimodularity.*
*In this case, it guarantees that the flow will be 0 or 1 at the solution, even if no constraint imposes it explicitly.*